

1 • Shared evidence: the rescue route

Heart route rescue

Name _____ Date _____

The cards describe a simplified circuit. It is a route map, not an anatomical picture.

Route A

Right heart → lungs. Blood is oxygen-poor on the way to the lungs.

Route C

Left heart → body. Blood delivers oxygen to tissues.

Vessel key

Artery = away from heart. Vein = towards heart. Capillaries allow exchange with tissues.

Route B

Lungs → left heart. Blood has gained oxygen in the lungs.

Route D

Body → right heart. Returning blood is oxygen-poor, not oxygen-free.

Pump and colour key

Heart muscle contracts; valves help prevent backflow. Human blood is red. Blue lines in models mean oxygen-poor.

W1. Put A-D in cycle order, starting with A. Where does blood gain oxygen?

W2. Repair “lungs straight to body” and “all veins contain blue blood”.

W3. Name the heart-to-lungs vessel type. Explain using direction.

2 • Supported route: follow the token

Use page 1. Words to use: right heart / lungs / left heart / body.

S1. Complete the route: BODY → _____ → _____ → _____ → BODY.

S2. Where does blood gain oxygen?

- A. Lungs
- B. Heart muscle

S3. A vessel carries blood away from the heart. It is an ...

- A. Artery
- B. Vein

**S4. Complete: Heart _____ contract to pump _____. Valves help prevent _____ flow.
Word bank: backward / muscles / blood.**

S5. Explain what blue means on a blood-flow diagram. What colour is blood?

S6. Draw the complete route with four arrows.

3 • Standard route: repair the map

Use page 1. Labels must remain clear if printed without colour.

N1. Draw a complete cycle using both heart sides, lungs and body. Mark where oxygen enters the blood.

Direction	Artery or vein?
Right heart → lungs	
Lungs → left heart	
Left heart → body	
Body → right heart	

N2. Complete the table and explain the rule you used.

N3. Explain the different jobs of heart muscle and valves.

N4. Repair: “The blue vessel must contain blue blood.”

4 • Stretch route: audit a teaching model

A good model must be accurate about the relationships it shows.

T1. Repair and redraw: right heart → body → lungs → left heart → right heart.

T2. Explain why an oxygen-poor artery is possible. Give the route.

T3. Describe how the heart muscle and valves work together. Which supplies the pumping force?

T4. Identify two limits of a four-box route model. Explain one useful detail to add.

5 • Exit and reflection

Use direction words, not just colours.

E1. Where does blood travel immediately after the lungs in our route?

E2. What defines an artery?

E3. What produces the force that pumps blood?

E4. What does blue mean in this model? Is blood blue?

Reflection. Which route error can you now explain?
